EARTH'S FIELD NMR

Jason Gallicchio and Claude, rev A, 2026-11-02

USING IT (docs/HANDBOOK.md has the whole story)
1 Wind the coil: 916 turns of AWG 20 in 4 layers on a 89 mm former, 200 mm long (26.8 mH, 8.9 ohm); fill it with 1 litre of water.
2 Power: +15 V and −15 V at J1; 6–30 V, up to 3 A, at POLARIZE SUPPLY (a battery is the quietest).
3 Tune SW1 to the protons where you are: f = 42.577 Hz/uT x B. Claremont, 46.0 uT: 1960 Hz, switches 1 2 4 7 ON.
   Earth's field runs 22–67 uT, 937–2853 Hz: SW1 reaches it everywhere with the coil above.
4 MODE RUN, SEQUENCE TIMER: RV1 sets how long it polarizes, RV2 how long it listens; or drive POLARIZE IN.
5 Watch OUT on a scope, triggered on SYNC OUT falling; CAL IN puts a known signal into the tank.
6 Or listen: headphones in PHONES, VOLUME clockwise louder. Start it low: OUT can hit the rails at a switch-off.
CAUTION: the coil and Q1's drain reach about 130 V for half a millisecond at every switch-off.
Copper: 1 (top) every drawn wire; 2 ground; 3 +15 V; 4 −15 V and the three hops.

https://github.com/gallicchio/earth-nmr-pcb
circuit, scripts, handbook

Earth's Field NMR rev A, 2026-11-02: bottom, seen from below (mirrored), scale 1.05:1